Day 029 | Unit 02 | Teacher plan

Positive Rational Numbers: Fractions and Mixed Numbers

Explain Fraction Models and Algorithms

Standards: 6.NS.A.1

Learning target: I can explain Fraction Models and Algorithms, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole. If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. Rename a mixed number as (whole × denominator + numerator)/denominator. Keep the dividend, change division to multiplication, and use the reciprocal of the divisor. Simplify and interpret the result. Identify the total amount and the size of one group. Use total ÷ group size for the number of groups. A quotient may include a partial group; if the question asks for full servings, use only the whole-number part.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 15

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01 / Learn

A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole. If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. Rename a mixed number as (whole × denominator + numerator)/denominator. Keep the dividend, change division to multiplication, and use the reciprocal of the divisor. Simplify and interpret the result. Identify the total amount and the size of one group. Use total ÷ group size for the number of groups. A quotient may include a partial group; if the question asks for full servings, use only the whole-number part.

Worked example 1

Draw a bar partitioned into 12 equal parts. Shade 6 parts. Circle groups of 3 parts. Use the model to find 1/2 ÷ 1/4.

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Rename both fractions with a common denominator, draw equal parts, and count groups.

3. Each small part is 1/12

4. 6 shaded parts divided into groups of 3 gives 6/3 = 2 groups.

5. State the result in the form requested: 2.

Answer: 2

Worked example 2

A jug contains 1/3 L. Each full bottle needs 1 L. How many full bottles can be filled, and how much liquid remains?

1. Identify the total amount and the size of one group.

2. 1/3 ÷ 1 = 1/3 × 1 = 1/3.

3. Only whole bottles count

4. subtract the liquid used.

Answer: 0 bottles; 1/3 L left

Day 029 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a bar partitioned into 4 equal parts. Shade 3 parts. Circle groups of 2 parts. Use the model to find 3/4 ÷ 1/2.

02 / Constructed response

Calculate 3/4 ÷ 1/2. Simplify.

03 / Constructed response

Calculate 1 3/4 ÷ 1/2. Simplify.

04 / Constructed response

A ribbon is 3/4 m long. Each piece is 5/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

05 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 2 parts. Circle groups of 2 parts. Use the model to find 1/4 ÷ 1/4.

06 / Constructed response

Calculate 1/4 ÷ 5/4. Simplify.

07 / Constructed response

Calculate 1 3/8 ÷ 2 2/3. Simplify and explain why your answer is reasonable.

08 / Model and constructed response

A jug contains 3/8 L. Each full bottle needs 1 L. How many full bottles can be filled, and how much liquid remains?

09 / Multiple choice

Draw a bar partitioned into 8 equal parts. Shade 5 parts. Circle groups of 4 parts. Use the model to find 5/8 ÷ 1/2. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 1/4
B. 25/2
C. 9/4
D. 5/4

10 / Constructed response

Calculate 2 ÷ 5/8. Simplify and explain why your answer is reasonable.

Day 029 | Unit 02 | Answer key

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1. 3/2

Each small part is 1/4; 3 shaded parts divided into groups of 2 gives 3/2 = 3/2 groups.

6.NS.A.1 | Constructed response

2. 3/2

3/4 ÷ 1/2 = 3/4 × 2 = 3/2.

6.NS.A.1 | Constructed response

3. 7/2

7/4 ÷ 1/2 = 7/4 × 2 = 7/2.

6.NS.A.1 | Constructed response

4. 3/5 piece-lengths

3/4 ÷ 5/4 = 3/4 × 4/5 = 3/5.

6.NS.A.1 | Constructed response

5. 1

Each small part is 1/8; 2 shaded parts divided into groups of 2 gives 2/2 = 1 groups.

6.NS.A.1 | Constructed response

6. 1/5

1/4 ÷ 5/4 = 1/4 × 4/5 = 1/5.

6.NS.A.1 | Constructed response

7. 33/64

11/8 ÷ 8/3 = 11/8 × 3/8 = 33/64.

6.NS.A.1 | Constructed response

8. 0 bottles; 3/8 L left

3/8 ÷ 1 = 3/8 × 1 = 3/8. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

9. D - 5/4

Each small part is 1/8; 5 shaded parts divided into groups of 4 gives 5/4 = 5/4 groups.

6.NS.A.1 | Multiple choice

10. 16/5

2 ÷ 5/8 = 2 × 8/5 = 16/5.

6.NS.A.1 | Constructed response